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# Requirements and models you can trust

Lecture 2 · CADE30008 Flight Dynamics & Advanced Control

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# How much of a problem is this?

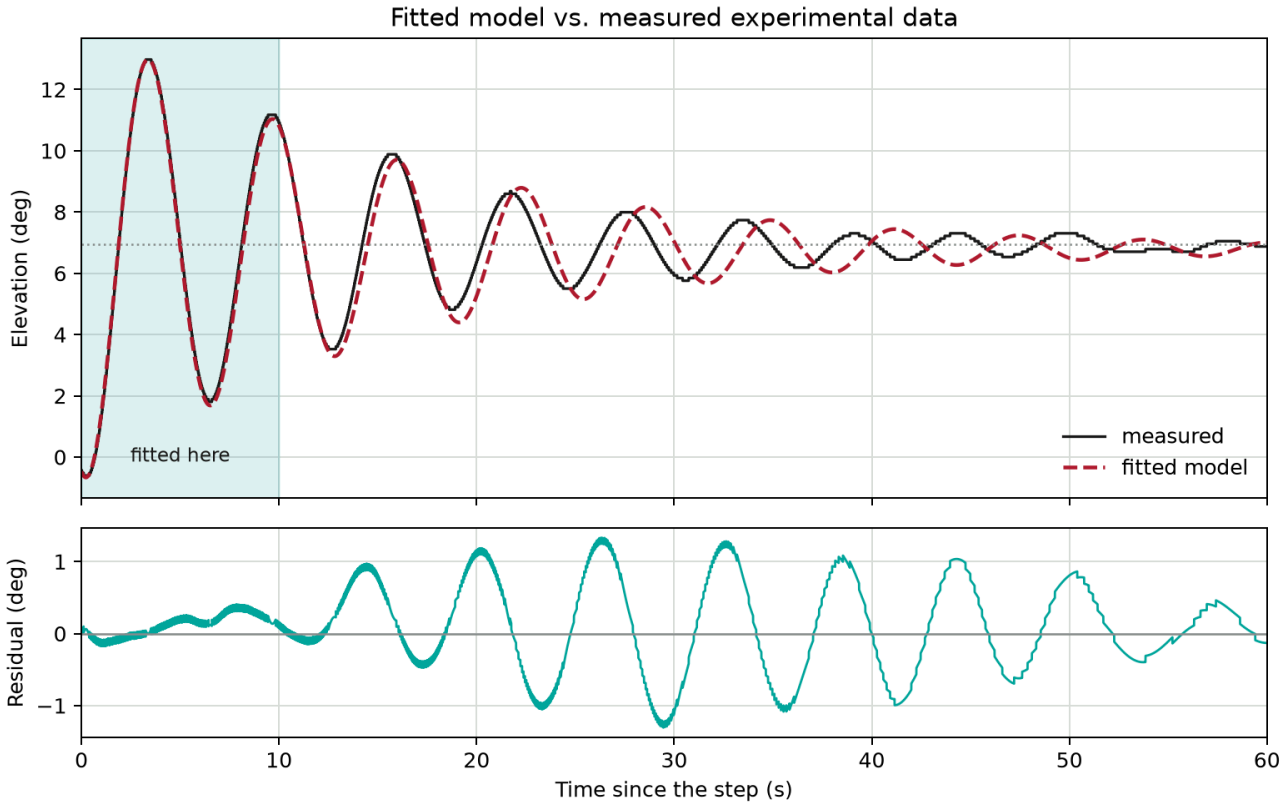
Hook and recall · 5 min

# Before anything new

1. You fitted K, $\omega_n$ and $\zeta$ from one recording. Which do you trust **most**, and which **least**?
2. A classmate's $\zeta$ differs from yours by 10%. Is one of you wrong?
3. Your design met the requirement in simulation. What would have to be true for it to meet it on the rig?

**Answer all three before any answer goes up.**

# The model, on top of the recording it came from




**How much of a problem is this?**

# It depends what you want it for

| Over the first | They differ by | Worst case |
|---|---|---|
| 10 s | **2.8%** of the 6.80 deg step | 0.41 deg |
| 60 s | **8.9%** | 1.33 deg |

**The model did not get worse. It was always a claim about a window.**

# One recording, one trim, one step size

- The arm was already swinging, through **4.21 deg**, when the step arrived
- `L02_validate` fits the starting state, which is the shaded window. So does MATLAB's `compare`
- Two minutes with `plot_recording` in the room would have saved it needing the repair
- Even from rest, a fit is a claim about **one trim elevation and one amplitude**

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# What do we want?

Learn A · 15 min

# A number, a test, and a reason

| Where the number came from | Example |
|---|---|
| The vehicle's job | A camera swath sets the tracking tolerance |
| A handling-qualities criterion | Closed-loop damping inside the standard's band |
| A physical limit you measured | The demand saturates, so cap the controller |
| An agreement in the room | "Eight seconds", for want of a reason |

**The last one is legitimate. Dressing it up as one of the first three is not.**

# A settling time is a frequency

$$\zeta = \frac{-\ln M_{\mathrm{p}}}{\sqrt{\pi^2 + \ln^2 M_{\mathrm{p}}}}, \qquad \omega_{\mathrm{n}} = \frac{4}{\zeta\, t_{\mathrm{s}}}$$

| Settle to 2% in | Overshoot at most | ζ | $\omega_n$ |
|---|---|---|---|
| 1.5 s | 10% | 0.591 | 4.51 rad/s |
| 2.0 s | 10% | 0.591 | 3.38 rad/s |
| 3.0 s | 10% | 0.591 | 2.26 rad/s |
| 4.0 s | 20% | 0.456 | 2.19 rad/s |

**Your model has to be right at that frequency.**

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# Make these testable

Do A · 10 min

YOU DO

# Three things somebody actually said

1. "It should feel crisp"
2. "It mustn't wander off while the camera is running"
3. "Don't let it fight the pilot"

> Rewrite each as a requirement. Name **which of the four sources** its number came from, and **one number you invented**.
>
> **In pairs, ten minutes.** Done when all three have a quantity and a number.

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# What do we know?

Learn B · 15 min

# A model is trustworthy when we know three things

- **Over what frequencies, or over what time.** 10 s, not 60 s
- **Over what amplitude, and about what trim**
- **How far out it is inside that**, from a residual

> **Not being right. Nothing is right.** A model handed over without these three is three coefficients and a hope.

# The residual's shape is the diagnosis

| What it looks like | What it is telling you |
|---|---|
| Flat, the size of the sensor noise | Nothing systematic left |
| A slow drift one way | The gain is out, or the trim moved |
| Oscillation at the signal's frequency, growing | The **frequency** is out |
| Oscillation at some **other** frequency | A **missing mode** |

**None of this is visible in a fit percentage.**

# A fit percentage hides a great deal

- **Fitted over what?** Match the starting state over 60 s instead of 10 s: the 60 s residual improves from 8.9% to 8.3%, and the 10 s residual doubles, 2.8% to 6.7%
- **Against what?** 5% of the change the input asked for, or 5% of the signal's own range, are different numbers

> Least squares spreads the error over whatever you hand it.

# "Near the trim" is a number

A linear model cannot, whatever its coefficients:

- **change its gain** with the size of the input
- **change its natural frequency** with the size of the motion

> So a measured dependence on amplitude is **outside the model by construction**, and no refit brings it inside.

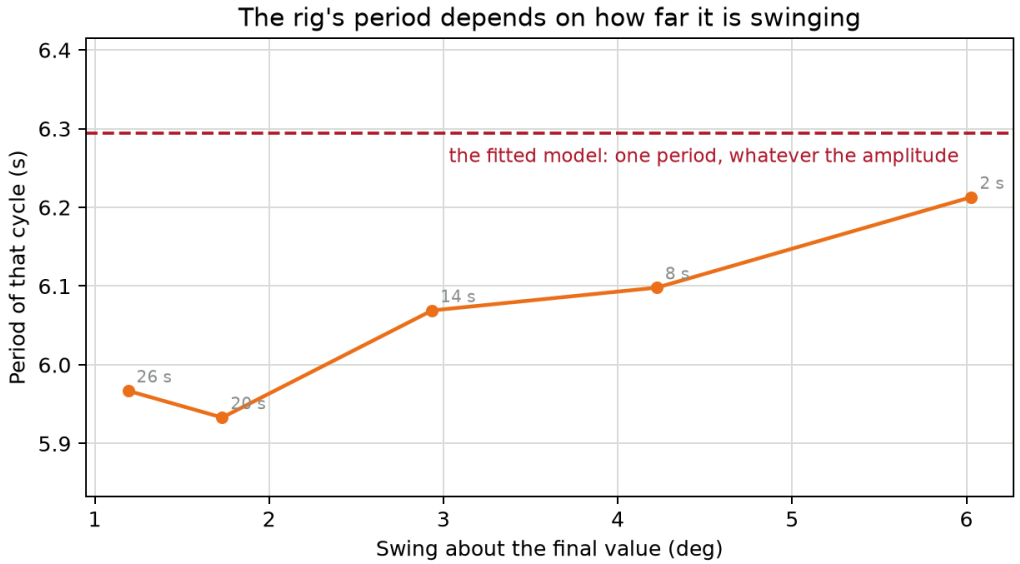
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# Where does it fit?

Do B · 10 min

YOU DO

# Measure the period, one cycle at a time




`L02_validate.m` · from **6.213 s** at **6.03 deg** of swing to **5.967 s** at **1.19 deg**

**In pairs, ten minutes.** Done when you can say the band of amplitudes your model is good for.

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# The case

Case brief · 5 min

# A pitch model, and two flight-test records

$$G(s) = \frac{\theta(s)}{\delta_{\mathrm{e}}(s)} = \frac{40}{s\,(s+2)\,(s+10)}$$

- Derived from geometry and aerodynamic data, and **handed to you**
- Two records, 60 s each, logged at 50 Hz: elevator, pitch rate, attitude
- Each flies a **3-2-1-1**, then a **pulse held and released**
- One at **2 degrees** of elevator, one at **8**

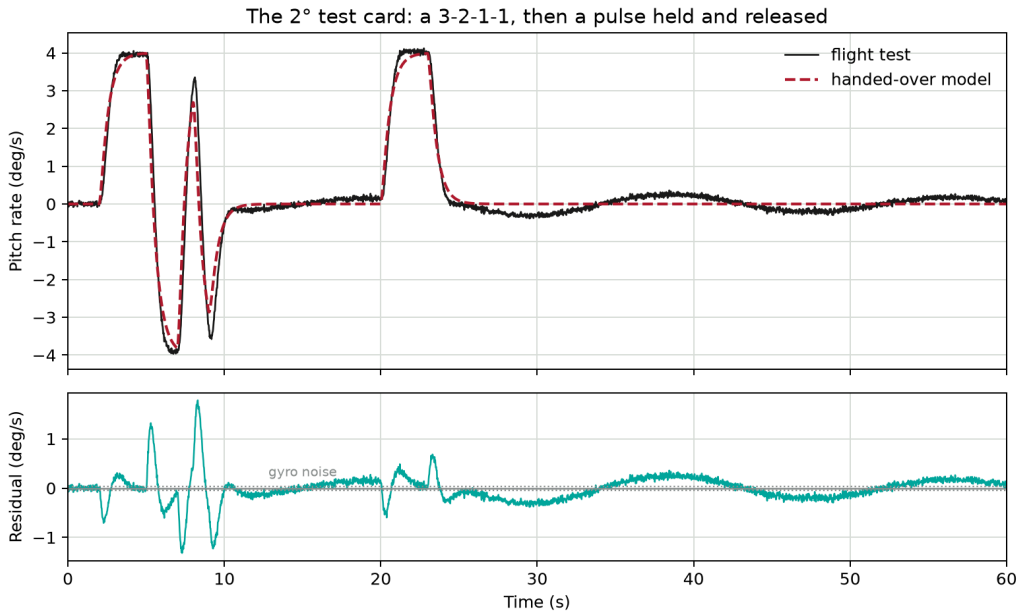
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# Validate it, bound it, write the requirement

Case build · 20 min

# Core expectation: simulate, subtract, look

YOU DO




Residual **14.2%** over the 3-2-1-1, **4.5%** after the release · gyro noise **1.0%**

**Pairs or threes, twenty minutes.** Done when you have the overlay, the residual, and the three observations.

# Observation 1: after the release, the record waves and the model is flat

- From 24 s the elevator is back at zero, and the model's rate decays to zero with it
- The record oscillates anyway: period **17.9 s**, so **0.35 rad/s**, about **0.28 deg/s**
- Two real poles and an integrator **cannot oscillate at any frequency**

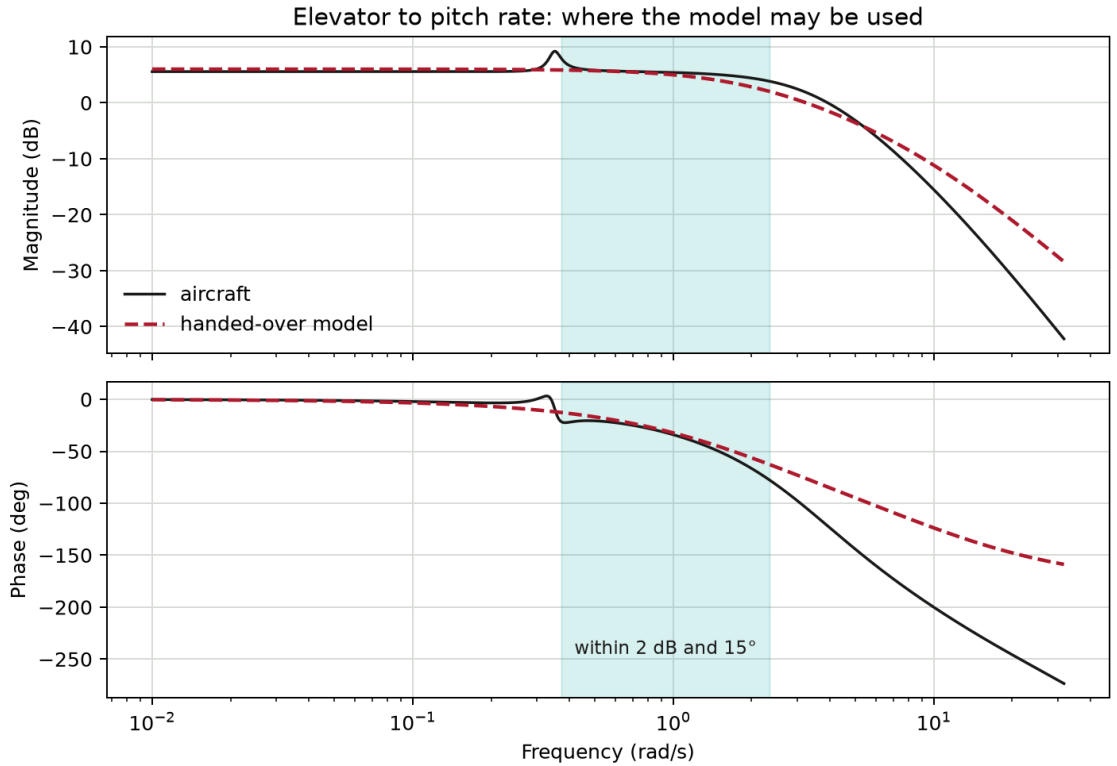
> **A residual with a frequency in it is a mode the model does not have.**

# Observation 2: not wrong, wrong above a frequency

| Everything below | 2 deg record | 8 deg record |
|---|---|---|
| 0.5 rad/s | 1.0% | 1.8% |
| 1.0 rad/s | 1.8% | 2.8% |
| 2.0 rad/s | 5.1% | 4.6% |
| 3.0 rad/s | 8.5% | 7.1% |
| 10 rad/s | 13.8% | 12.0% |

**That is a sentence a designer can act on.**

# What the aircraft actually was




Agreement within 2 dB and 15° from **0.37** to **2.35 rad/s**

# Observation 3: one gain in the model, two in the data

| Elevator | Pitch rate per degree |
|---|---|
| 2 degrees | **1.980** (deg/s)/deg |
| 8 degrees | **1.798** (deg/s)/deg |
| The model, at any amplitude | 2.000 (deg/s)/deg |

**A surface at a large deflection is less effective than the same surface at a small one.**

# Extension: can I ask my model to do this?

- **Settle to 2% in 3.0 s**, overshoot at most 10% → $\omega_n$ = **2.26 rad/s**, inside the band
- **Settle to 2% in 2.0 s → 3.38 rad/s**, outside it

> Relax the requirement, or go and fly more test points. **Designing there anyway and reporting the simulation is not an option.**

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# What the room found

Converge · 10 min

# Three observations and one argument

- **The bound.** Between the slow mode and about 2 rad/s, a few degrees of elevator, around 5% inside that
- **The missing mode**, at 0.35 rad/s
- **The amplitude.** One gain in the model, two in the data

> **Is this model good enough?** It depends entirely on the requirement.

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# What this cannot do

Cliffhanger · 10 min

# You have both halves, and it still does not work

- A model you can defend, with its range written down
- A requirement that can be met, missed or argued about
- A linear analysis that is **correct**

**And the response goes a long way past where it was asked to stop.**

# Today, in five lines

- A model is a **claim with a range of validity**: frequencies, amplitude, and how far out inside that
- **Read the residual's shape**, not its size
- **Quote the window.** Same model, same recording: 2.8% over 10 s, 8.9% over 60 s
- **Linearisation is about a trim and an amplitude.** The rig's period moved 4%; the aircraft's gain moved 9%
- A requirement names a quantity, a number, a test **and a source**

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# A model you trust and a requirement you can test

How to use them properly?

See you next week